



SUSTAINABLE OLYMPIAD

# Sustainable Olympiad 2025

## Round 1: Online Knowledge Challenge

### Junior Division (ages 12-14)

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|---------------------|----------------------------------------|
| <b>Time allowed</b> | 90 minutes                             |
| <b>Total marks</b>  | 50                                     |
| <b>Part A</b>       | 10 multiple-choice questions, 20 marks |
| <b>Part B</b>       | 4 structured problems, 30 marks        |

## Instructions

- Answer **all** questions. Work together as a team.
- Part A: choose **one** answer (A, B, C or D) for each question. There are no negative marks.
- Part B: show all your working. Marks are given for correct method even if the final answer is wrong.
- Give numerical answers to an appropriate number of significant figures, with units.
- Scientific calculators are allowed. Internet access and AI tools are not allowed during the round.
- A data sheet is provided on the next page.

**Team name**

**School**

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## Data sheet

- 1 kWh = 1000 Wh = 3.6 MJ; 1 tonne (t) = 1000 kg
- 1 litre of water has a mass of 1 kg
- Heating 1 kg of water by 1 °C needs 4.18 kJ
- 1 hectare (ha) = 10 000 m<sup>2</sup>; 1 km<sup>2</sup> = 100 ha
- Electricity: 0.25 kg CO<sub>2</sub> per kWh (where needed)

## Part A: Multiple choice (20 marks)

Each question is worth 2 marks. Choose one answer.

**A1.** A classroom replaces a 60 W filament bulb with an 8 W LED bulb that gives the same light. The light is on for 5 hours a day, 365 days a year. How much energy is saved in a year? **[2]**

- A** 9.5 kWh
- B** 94.9 kWh
- C** 109.5 kWh
- D** 949 kWh

**A2.** Which energy source does NOT originally come from sunlight? **[2]**

- A** Wind
- B** Hydroelectric
- C** Wood
- D** Tidal

**A3.** In photosynthesis,  $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ . How many molecules of oxygen are released for each molecule of glucose made? **[2]**

- A** 1
- B** 3
- C** 6
- D** 12

**A4.** In the food chain grass → grasshopper → frog → snake → hawk, the grass stores 100 000 kJ of energy. If 10% of the energy passes from each level to the next, how much reaches the hawk? **[2]**

- A** 1000 kJ
- B** 100 kJ
- C** 10 kJ
- D** 1 kJ

**A5.** Making a can from recycled aluminium instead of from bauxite ore saves about how much of the energy? **[2]**

- A** About 10%
- B** About 40%
- C** About 70%
- D** About 95%

- A6.** Which greenhouse gas is the most abundant in Earth's atmosphere? [2]
- A** Carbon dioxide
  - B** Methane
  - C** Water vapour
  - D** Ozone
- A7.** Seawater has a density of about  $1.025 \text{ g cm}^{-3}$ . Which plastic will float in the sea? [2]
- A** PET, a drinks bottle ( $1.38 \text{ g cm}^{-3}$ )
  - B** PVC, a pipe ( $1.38 \text{ g cm}^{-3}$ )
  - C** Polypropylene, a bottle cap ( $0.90 \text{ g cm}^{-3}$ )
  - D** Solid polystyrene, a CD case ( $1.05 \text{ g cm}^{-3}$ )
- A8.** A flight emits  $0.15 \text{ kg CO}_2$  per passenger per km. Lisbon to Paris is 1450 km. How much  $\text{CO}_2$  is emitted for one passenger making a **return** trip? [2]
- A** 43.5 kg
  - B** 217.5 kg
  - C** 435 kg
  - D** 4350 kg
- A9.** Two meadows each contain 100 plants of the same 4 species. Meadow X has 25 of each species. Meadow Y has 97 of one species and 1 of each of the others. Which statement is correct? [2]
- A** Both have the same biodiversity because they have the same number of species
  - B** Meadow X has higher biodiversity because the species are more evenly spread
  - C** Meadow Y has higher biodiversity because one species is very successful
  - D** Biodiversity cannot be compared without knowing the area
- A10.** Why do tropical rainforest soils often become poor for farming only a few years after the forest is cleared? [2]
- A** Tropical soils contain too much nitrogen
  - B** Most nutrients are stored in the living plants, and heavy rain washes the rest out of the thin soil
  - C** The soil becomes too cold without the trees
  - D** Earthworms eat all the organic matter

## Part B: Structured problems (30 marks)

### B1. Computers left on (8 marks)

A computer room has 12 computers. Each uses 120 W while in use and 60 W while idle. They are used for 6 hours each school day but left idle for the other 18 hours instead of being switched off. There are 190 school days a year. Electricity costs €0.20 per kWh, and each kWh causes 0.25 kg of CO<sub>2</sub>.

**(a)** Calculate the energy wasted per school day, in kWh. [2]

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**(b)** Calculate the energy wasted per school year. [2]

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**(c)** How much money is wasted per year? [2]

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**(d)** How much CO<sub>2</sub> is caused by the wasted energy each year? [2]

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### B2. From food waste to compost (8 marks)

A school canteen throws away 15 kg of food waste every school day (190 days a year). In a landfill, each kg of food waste releases 0.05 kg of methane. Methane traps 28 times as much heat as the same mass of CO<sub>2</sub> over 100 years. Composting reduces the mass of the waste by 60%.

**(a)** Calculate the total food waste per year. [1]

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**(b)** What mass of compost would be produced? [2]

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**(c)** Calculate the CO<sub>2</sub>-equivalent emissions avoided each year by composting instead of using a landfill. Ignore any emissions from the compost heap. **[3]**

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**(d)** Explain why a well-managed compost heap releases much less methane than a landfill. **[2]**

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### **B3. Cycling to school (7 marks)**

Ana lives 4.0 km from school. Each school day she is driven to school and back home (8.0 km in total). The car emits 120 g of CO<sub>2</sub> per km. There are 190 school days a year.

**(a)** Calculate the CO<sub>2</sub> emitted by these journeys in a year, in kg. **[2]**

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**(b)** 30 students with the same journey start cycling instead. How much CO<sub>2</sub> do they save in total? **[2]**

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**(c)** The car averages 30 km/h and Ana cycles at 15 km/h. How many extra minutes per day does cycling take? **[3]**

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### **B4. Reading climate data (7 marks)**

Average CO<sub>2</sub> concentration measured at Mauna Loa, Hawaii: 1960, 317 ppm; 1980, 339 ppm; 2000, 370 ppm; 2020, 414 ppm.

**(a)** Calculate the average rate of increase, in ppm per year, for each 20-year period. **[3]**

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**(b)** Describe what your answers show. **[2]**

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**(c)** Predict the concentration in 2030 if the 2000–2020 rate continues. **[2]**

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**END OF PAPER**